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A SIMPLE PHYSICAL PRINCIPLE FOR THE SIMULATION OF FREEWAYS WITH SPECIAL LANES AND PRIORITY VEHICLES

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Abstract—This paper presents a simple physical principle that can be used to solve the kinematic wave problem for freeways with special lanes and priority vehicles. The principle is shown to yield the flows for all possible 'Riemann problems' arising in a homogeneous highway, so that its application in a simulation is equivalent to the Godunov 'classic' finite difference approximation method. The principle is appealing because its physical basis, unlike purely mathematical formulae, suggests a natural way in which boundary conditions for practical problems may be treated. Perhaps the IT principle will prove useful for solving general problems, e.g. involving multi-commodity networks. This issue deserves more study. As an illustration of this potential the paper shows that an IT simulation of the finite highway problem solved in the companion paper (Daganzo, *Transportation Research B*, **31**, 83–102, 1997) matches rather well the exact solution. Additional tests using other boundary conditions for the same problem also revealed a good match. © 1997 Elsevier Science Ltd. All rights reserved

1. BACKGROUND

The subject of this paper is a long crowded freeway with two classes of vehicles and two sets of lanes, a set of regular lanes and a set of special lanes that only one of the vehicle types (priority) is allowed to use. An example of such a facility is a freeway with a carpool lane. Another example is a freeway immediately upstream of a congested off-ramp, where the exiting vehicles naturally avoid the 'far-side' lanes. Through vehicles can then be viewed as 'priority' vehicles that are entitled to use the far side lanes. A simple continuum theory of traffic flow on such facilities already exists. The purpose of this paper is to demonstrate that there is a very simple and physically intuitive way of simulating traffic that is equivalent to the finite difference approximation results presented in Daganzo (1997). In order to make the presentation self-contained, the next two subsections summarize the pertinent results in that reference. The new material begins in Section 2.

1.1. The Continuum Model for Freeways with Priority Vehicles

Let $u(k)$ denote the steady state, non-increasing, speed–density relation that would prevail on our freeway if it did not have special lanes, and assume that the companion flow–density relation, $ku(k)$, is of the simple triangular form depicted in Fig. 1. This simple form is attractive because $ku(k)$ can then be specified in terms of just three interpretable parameters: the slopes of the two lines (v_f and $-w$) and the maximum flow, $q_{\max}$. The slope v_f denotes the speed of unqueued traffic and the slope $-w$ the (negative) velocity with which disturbances in speed propagate within a queue. Alternatively, instead of one of these parameters, one can use the 'jam' or maximum possible density, $k_{\max}$, since we see from the figure that $k_{\max} = (q_{\max}/v_f) + (q_{\max}/w)$.

Figure 1 also shows that $ku(k)$ can be expressed as the lower envelope of a non-decreasing ('sending') function $S_T(k)$ and a non-increasing ('receiving') function $R_T(k)$. These functions intersect at the 'optimum' density, k_o, and take values in the $[0, q_{\max}]$ interval for k ranging from 0 to $k_{\max}$.

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In order to capture the special nature of our freeway we assume, as a first approximation, that the (space) mean speeds that would prevail in each set of lanes in a steady state with no lane-changing are only a function of the vehicular density in each set of lanes, k_r and k_s. This condition can be expressed more precisely if we introduce F and f as the fractions of freeway width that are reserved respectively for priority and regular vehicles ($F+f=1$). Then, we can write:

$$v = u(k_r/f) \text{ and } V = u(k_s/F); \tag{1}$$

this says that the speed prevailing in each set of lanes is the same as the speed that would prevail on the freeway as a whole if it was filled with the same vehicular density per lane.

It also stands to reason that in a steady state the priority vehicles would choose the set of lanes that move faster and that, as a result, only steady states with $v \leqslant V$ can arise. If we let k and K denote the density of each vehicle type (with $K_T = K + k$ being the total density), this means that the following Wardropian conditions must be true:

$$v \leqslant V, \tag{2a}$$

$$k_s = K \qquad \text{if } v < V, \text{ and} \tag{2b}$$

$$V = v = u(K_T) \qquad \text{if } v = V. \tag{2c}$$

The situation of eqn (2b) (with no priority vehicles on the regular lanes) is termed a 2-pipe regime because, as explained in Daganzo (1997), in that case the two sets of lanes flow like independent pipes. Conversely, eqn (2c) describes a 1-pipe regime. Note that in a 2-pipe regime the lane speeds, V and v, are also the speeds of the two vehicle types, and that vehicles of the same type travel at the same speed. Since the same is true for the 1-pipe regime, it is appropriate to think of $\mathbf{V} = (V, v)$ as the speed vector by lane and/or by vehicle type.

It also follows from eqns (1) and (2) that an arbitrary density vector by vehicle type ($\mathbf{K} = (K,k)$) uniquely determines the set of lane/vehicle speeds that can prevail in the steady state. In particular, we know that if $(K/F) < (k/f)$ then the system must be in the 2-pipe regime because even when all the priority vehicles are put on the special lanes, i.e. $k_r = k$ and $k_s = K$, these lanes are less crowded and exhibit a higher speed; therefore:

$$v = u(k/f) \text{ and } V = u(K/F), \qquad \text{if } (K/F) < (k/f). \tag{3a}$$

If on the other hand, $(K/F) > (k/f)$, we see from eqn (1) that a full allocation of priority vehicles to the special lanes would result in lower speeds on these lanes. Clearly, the only

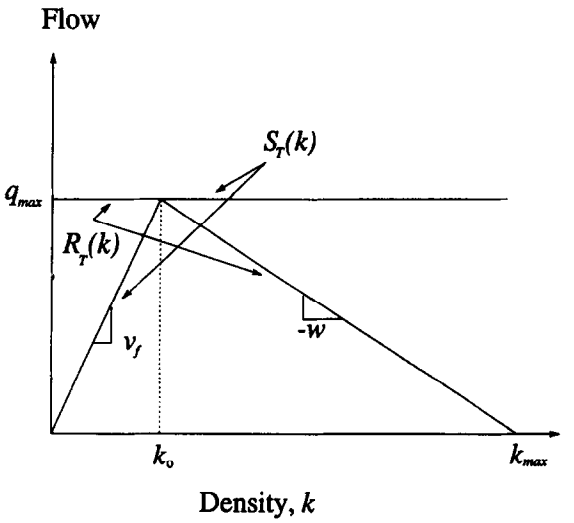


Fig. 1. A triangular flow–density relation and its parameters.

Wardropian equilibrium arises when priority vehicles distribute themselves across lanes so as to make the speed on both sets of lanes equal. No vehicle would then have an incentive to change lanes. This observation leads to the condition:

$$v = V = u(K + k), \qquad \text{if } (K/F) > (k/f). \tag{3b}$$

Note that if $K/F = k/f$, then eqn (3) yield the same answer; either expression can be used.

We also note that the equilibrium vehicular distribution across lanes must be:

$$k_r = k \text{ and } k_s = K, \qquad \text{if } (K/F) \leqslant (k/f) \qquad \text{and} \tag{4a}$$

$$k_r = f(K + k) \text{ and } k_s = F(K + k), \qquad \text{if } (K/F) \geqslant (k/f). \tag{4b}$$

The macroscopic theory in Daganzo (1997) assumes that a crowded freeway can be modeled as a density continuum over time (t) and space (x), $\mathbf{K}(t,x)$, that satisfies the conservation of mass equation and the relations embodied in eqn (3), i.e. $\mathbf{V}(t, x) = \mathbf{V}(\mathbf{K}(t, x))$, at all points. The theory is intended to model macroscopic phenomena on long freeways such as the development and spatial evolution of separate vs coalesced queues on the two sets of lanes, treating the transitions between queueing states (e.g. those involving lane-changing) as 'shocks' that are not explicitly modeled in the theory. This approach should be reasonable if these transitions take place over relatively short lengths of freeway. (Because lane-changing driver interactions are not well understood it is appealing to leave them out in a macroscopic theory.) Of course, experimental comparisons should help determine whether such an approximation is appropriate. The results in this paper should make it easy to computerize the theory and to generate numerical predictions that can be compared against field data.

In what follows it will be necessary to talk about the flows of the two vehicle types, $\mathbf{Q} = (Q,q)$. Such flows are related to the vehicular densities and (space) mean speeds by the relations: $q = kv$ and $Q = KV$. Together with eqn (3), these expressions induce the following vector-valued flow–density relation:

$$\mathbf{Q} = \mathbf{Q}(\mathbf{K}) = (KV(\mathbf{K}), kv(\mathbf{K})). \tag{5}$$

This relation is continuous. As explained in Daganzo (1997), it is of a form that exhibits constant wave speeds in each of the four regions of **K**-space (A, B, C, D) shown in Fig. 2.

It is also useful to let $M = Fq_{\max}$ and $m = fq_{\max}$ denote the maximum flows that are possible in the special and regular lanes, and $K_j = Fk_{\max}$ and $k_j = fk_{\max}$ denote the respective 'jam' densities. We see then that the slanted line passing through the origin of Fig. 2 and separating the 1-pipe and 2-pipe regimes as per the inequalities governing eqn (3), is given by the equation $k/K = f/F \equiv k_j/K_j \equiv m/M$. The four regions of constant

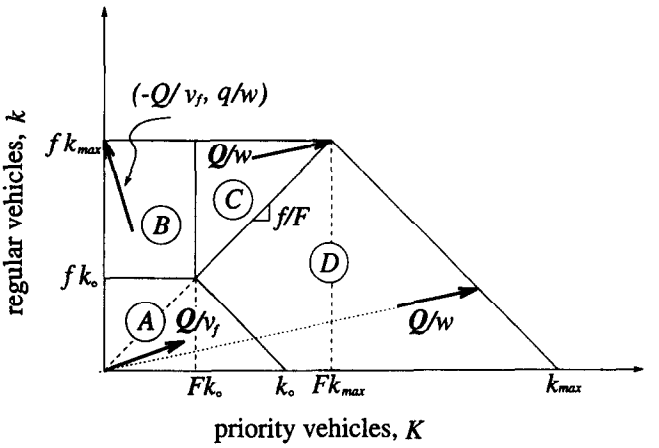


Fig. 2. The flow–density relation, expressed as a vector field.

wave speeds are: 'A' or 'free-flow' where both vehicle types travel at v_f, 'B' or 'semi-congested' where priority vehicles move at speed v_f and regular vehicles at a speed $v < v_f$, 'C' or '2-pipe, congested' where $v < v_f$, and 'D' or '1-pipe, congested' where $v = V < v_f$.

The most convenient way of expressing the flows in each of these regions, i. e. specifying eqn (5), is by means of an associated vector field in the (K,k)-plane, as shown in Fig. 2. In region 'A' the vector displayed is $\mathbf{Q}/v_f$; it points from the origin to the point in question, thus expressing the stated relation: $(Q, q) = (v_f K, v_f k)$. In region 'B' the vector displayed is $(-Q/v_f, q/w)$; it points from the point in question to the upper left corner of the region. In regions 'C' and 'D' the vector $(\mathbf{Q}/w)$ is based at the point in question. In region 'C' it points to the upper right corner of the region, and in region 'D' to the region boundary on a line passing through the origin (with slope k/K).

The easiest way to see why these representations are true is by examining from the figure how $\mathbf{Q}$ changes when K or k are changed a little, while holding the other density constant; i.e. by obtaining graphically the partial derivatives of $\mathbf{Q}$ with respect to K and k in each of the four cases. For example, in region 'C' we see that Q (or q) is independent of k (or K), and that Q (or q) decreases by w units when K (or k) increases by 1 unit. This, of course, is what one would expect from a state 'C' that behaves like two independent and queued freeways.

1.2. The Finite Difference Approximation

In the proposed finite difference approximation, it is assumed that the freeway has been partitioned into small sections (cells), i, of length δ. The averages of $K(t,x)$, $k(t,x)$ and $\mathbf{K}(t,x)$ in each one of these cells are then denoted by $K^i(t)$, $k^i(t)$ and $\mathbf{K}^i(t)$; note that these averages represent the number of vehicles (by type) in cell i at time t, divided by δ.

The proposed procedure revises the $\mathbf{K}^i(t)$ every ϵ time units with a 'conservative' rule:

$$\mathbf{K}^i(t+\epsilon) = \mathbf{K}^i(t) + (\epsilon/\delta)[\mathbf{Q}^{i-1}(t) - \mathbf{Q}^i(t)] \tag{6}$$

where $\mathbf{Q}^i(t)$ is the average flow moved out of section i during $(t,t+\epsilon)$. Note that $\epsilon\mathbf{Q}^i(t)$ represents the number of vehicles that are advanced from cell i to cell $i+1$ in the time interval from t to $t+\epsilon$.

A possible approach (Godunov, 1961) consists in choosing for $\mathbf{Q}^i$ the flows that would arise in an initial value problem where the densities upstream and downstream of the interface between cells i and $i+1$ (denoted $\mathbf{K}^u$ and $\mathbf{K}^d$) are constant and equal to those of the cells in question; i.e. $\mathbf{K}^u = \mathbf{K}^i(t)$ and $\mathbf{K}^d = \mathbf{K}^{i+1}(t)$.* For applications where these simple (Riemann) problems can be solved in closed form, Godunov's method is attractive; it has proven to be efficient and successful. Formulae for the Godunov flows, $\mathbf{Q}_G$, as a function of $\mathbf{K}^u$ and $\mathbf{K}^d$ for the model of Section 1 are given below.

The expressions can be written compactly in terms of the sending and receiving functions that define the flow–density relation for each set of lanes in the 2-pipe regime. Consideration of eqn (1) reveals that these functions are related to the sending and receiving functions of the whole freeway (see Fig. 1) by:

$$s(k_r) = f S_T(k_r/f), \quad r(k_r) = f R_T(k_r/f) \text{ for the regular lanes}$$

and

$$S(k_s) = F S_T(k_s/F), \quad R(k_s) = F R_T(k_s/F) \text{ for the special lanes.}$$

We shall state first the formulae for the case where the upstream state is in the 1-pipe regime and the downstream state in a congested or semi-congested 2-pipe regime. It was shown in Daganzo (1997) that the Godunov flows Q, q and Q_T (the subscript 'G' is dropped now for simplicity of notation) are:

*For the approach to work the time step ϵ should not exceed the cell size to maximum wave speed ratio; in our traffic application this condition is $\epsilon \leqslant \delta/v_f$, and the value $\epsilon = \delta/v_f$ is recommended.

$$Q_T = r(k^d) + R(K^d) \tag{7a}$$

$$q = \min\{r(k^d);\ Q_T k^u / K_T^u\} \tag{7b}$$

$$Q = Q_T - q \tag{7c}$$

if the upstream (1-pipe) state is congested. Equation (7) also applies if the upstream state is uncongested and to the right of the dashed line of Fig. 3 (i.e. 'subcase 1' in the terminology used in Daganzo, 1997) provided that $S_T(K_T^u) > r(k^d) + R(K^d)$. If this inequality is not satisfied, still in subcase 1, then

$$\mathbf{Q} = \mathbf{Q}^u. \tag{8}$$

Finally, if we are to the left of the dashed line ('subcase 2') or on it, the result is:

$$q = \min\{s(k^u);\ r(k^d)\} \tag{9a}$$

and

$$Q = \min\{S(K^u);\ R(K^d)\}. \tag{9b}$$

It turns out that eqn (9) is also the result for transitions between 2-pipe regimes. The result for transitions between 2-pipe (upstream) and 1-pipe (downstream) also follows eqn (9), provided that $\mathbf{K}^d$ is replaced by (0,0) if $\mathbf{K}^d$ is in region 'A' and by $(K^d + k^d)(M, m)/(M + m)$ if it is in region 'D'; this yields:

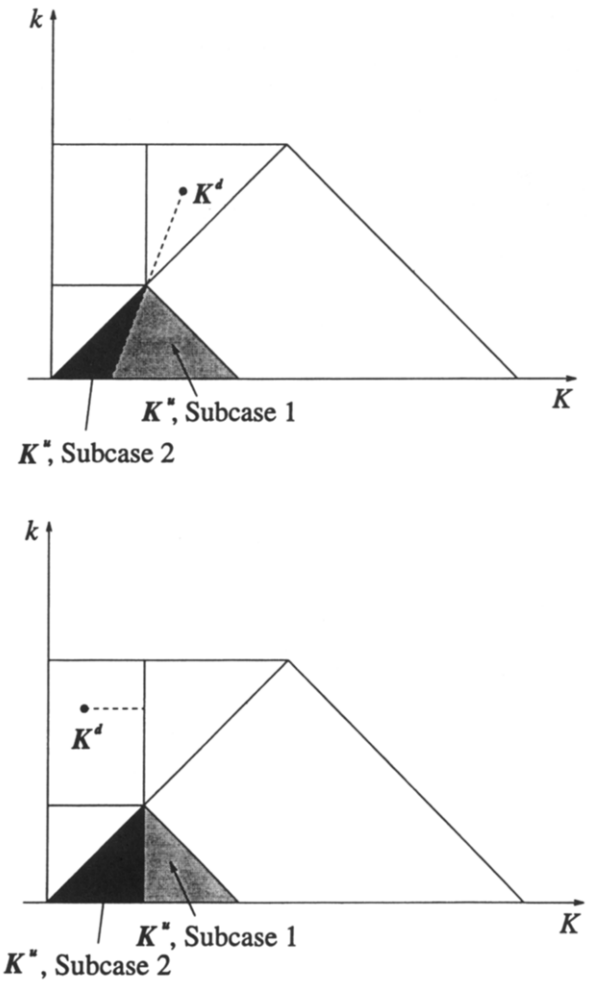


Fig. 3. Transitions from 1-pipe (uncongested) state to a 2-pipe (congested) state: two subcases.

$$q = s(k^u),\ Q = S(K^u) \text{ if } \mathbf{K}^d \text{ is in region A} \tag{10a}$$

and

$$\left.\begin{aligned} Q &= \min\{S(K^u);\ R((K^d + k^d)M/(M+m))\} \\ q &= \min\{s(k^u);\ r((K^d + k^d)m/(M+m))\} \end{aligned}\right] \text{ if } \mathbf{K}^d \text{ is in region D} \tag{10b}$$

The result for transitions between 1-pipe regimes is also easily stated:

$$(Q, q) = (K^u/K_T^u,\ k^u/K_T^u)\min\{S_T(K^u + k^u);\ R_T(K^d + k^d)\}. \tag{11}$$

This formula also applies if $\mathbf{K}^d$ is uncongested and in the 2-pipe regime.

Although eqns (7)–(11) are algebraically simple, they do not reveal a clear physical relationship between $\mathbf{Q}$ and $(\mathbf{K}^u, \mathbf{K}^d)$; the issue is explored in the remainder of this paper.

Section 2, below, introduces an 'incremental transfer' (IT) principle that can be put at the core of cell-transmission (CT) simulations. Section 3 proves that the simulated flows between two cells always match eqns (7)–(11). As a result, incremental transfer simulations can be expected to converge to the exact solution in practical problems. This is demonstrated by means of a collection of simple examples that includes all possible pairs of initial regimes. Section 4 discusses the treatment of boundary conditions for the non-homogeneous freeway problem and compares the simulated and exact solutions of a problem involving the spill-back of an off-ramp queue.

2. THE INCREMENTAL TRANSFER MODEL

First, let us define a set of 4 sending and receiving flow values, $S^i(t)$, $s^i(t)$, $R^i(t)$ and $r^i(t)$, for each cell i and each (lattice) time, t. The sending values $[S^i(t),\ s^i(t)]$ are intended to represent the number of vehicles of each type that the cell can emit per unit time, and the receiving values $[R^i(t),\ r^i(t)]$ the number of vehicles (type irrelevant) that the special and regular lanes can accommodate. These 4 values are related to the $\mathbf{K}^i(t)$ by the formulas given below. Explicit references to 'i' and 't' have been eliminated from the notation.

If $\mathbf{K}$ is in the 1-pipe case, then we define the quantities S, s, R and r as follows:

$$(S, s) = S_T(K+k)(K, k)/(K+k) \qquad \text{for } K/k > M/m \tag{12a}$$

and

$$(R, r) = R_T(K+k)(M, m)/(M+m) \qquad \text{for } K/k > M/m. \tag{12b}$$

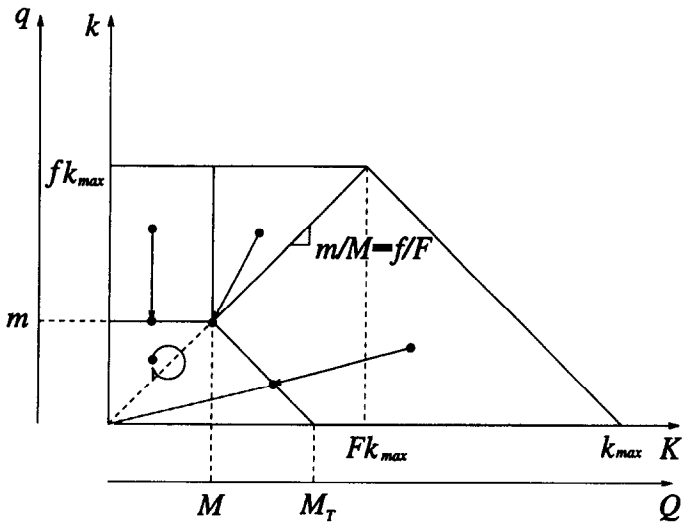


Fig. 4. Density-sending flow mapping.

If **K** is in the 2-pipe case, then

$$(S, s) = (S(K), s(k)) \qquad \text{for } K/k < M/m \tag{13a}$$

and

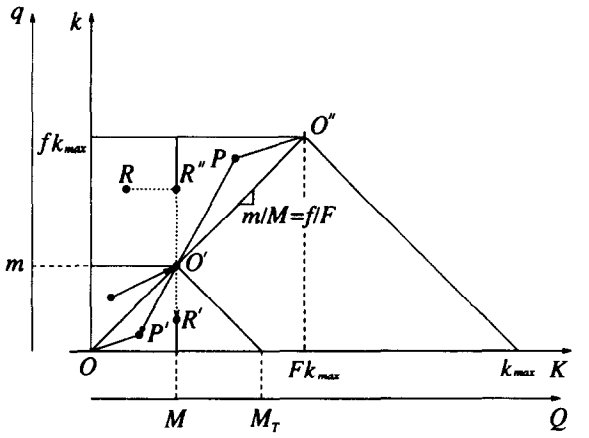
$$(R, r) = (R(K), r(k)) \qquad \text{for } K/k < M/m. \tag{13b}$$

If K/k = M/m, eqns (12) and (13) are equivalent; either can be used.

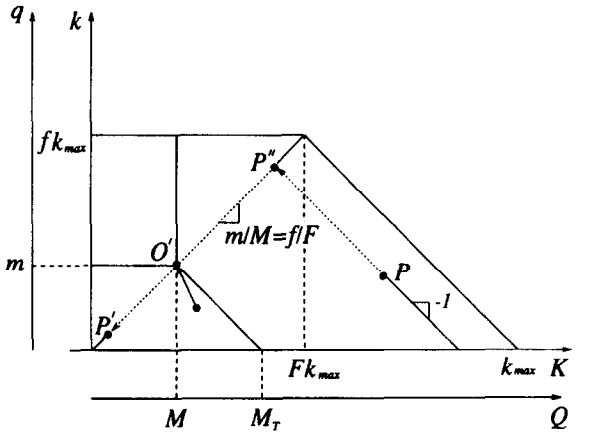
Equation (12a) indicates that the total flow that can be sent in the 1-pipe regime is the same as for a freeway without special lanes, $S_T(K_T)$, and that the fraction of this quantity corresponding to each vehicle class is proportional to the density. This means that the cell behaves as a regular freeway cell with no discrimination across vehicle classes. The same is true for the receiving flows since [eqn 12(b)] indicates that $R + r = R_T(K_T)$. Logically, the allocation of receiving 'room' to lane types only depends on the number of lanes of each type; i.e. on (F,f) or equivalently on (M,m).

The values (S,R) obtained from eqn (13) are those which would arise for an imaginary freeway consisting of the special lanes and containing the priority vehicles. Likewise, the values (s,r) are those of a separate freeway consisting of the regular lanes and containing the regular vehicles.

Figures 4 and 5 depict graphically the mappings induced by eqns (12) and (13) of **K** into the space of (S,s) and (R,r), i.e. into the (Q,q)-plane also shown in the figures. Arrows are used to point from a density to its corresponding flow. The diagram of Fig. 4 describes



(a) 2-pipe densities



(b) 1-pipe densities

Fig. 5. Density-receiving flow mapping. (a) 1-pipe regime; (b) 2- pipe regime.

the mapping completely and needs no discussion. This is not so in some of the cases in Fig. 5. We see that for points P and R, the arrows do not identify the flows unless something is said about their lengths. The details are given in the following paragraph.

The line used to identify point P′ from point P in Fig. 5a is straight. Furthermore, triangles OP′O′ and O″PO′ have parallel sides and are similar with sides in the ratio: w/v_f. These two aspects of the geometry of the figure, which follow from eqns (12) and (13), unambiguously identify a point P′ from any 2-pipe congested state such as P. It also turns out from eqns (12) and (13) that the position of any point R′ corresponding to a semi-congested state such as R can still be obtained with the same recipe, if it is applied to an auxiliary point R″ on the boundary of the semi-congested and congested regions with the same k. And likewise, the point P′ corresponding to an arbitrary 1-pipe, congested state such as P on Fig. 5b can still be identified in the same way, if the recipe is applied to an auxiliary point P″ on the boundary between regimes that lies on the line with slope -1 passing through P.

Note from Fig. 4 that the ratio of sending flows satisfies $S/s > \leqslant M/m \equiv F/f$ if and only if $K/k > \leqslant M/m$. Note as well from Fig. 5 that the receiving flow ratios always satisfy $R/r \geqslant M/m$, with the equality applying when $K/k \geqslant M/m$ (i.e. in the 1-pipe case). This is reasonable since, as one may expect, the amount of receiving room per lane in the special lanes should always be greater than or equal to that in the regular lanes.

2.1. The IT Principle

We now consider a recipe for determining the advancing flows between two neighboring cells. In order to avoid the proliferation of subscripts, S and s will refer from now on to the sending values of the upstream cell $(i-1)$ at time t, and R and r to the receiving values of the downstream cell (i) at t.

We imagine that the sending flows represent the component amounts of a fluid mixture placed in a reservoir that represents the sending cell* (see Fig. 6a). Directly below we have two reservoirs, representing the special and regular lanes of the receiving cell, that can accept amounts R and r of this imaginary fluid. The R-reservoir is M units wide and the r-reservoir m units wide. Both have the same height. The incremental transfer principle can now be stated.

When the spigot is open, the downstream reservoirs are filled gradually by placing regular (s-) fluid elements in the r-reservoir and priority (S-) fluid elements in the reservoir that is currently filled to the lesser height (i.e. less congested). In this spirit, the arrows in the figure denote possible flow transfers, with the dotted arrow being currently inactive. (Since $R/r \geqslant M/m$, we see that, initially, the R-reservoir is filled to a lesser or equal height than the r-reservoir.) The process continues until each fluid has been either completely transferred or blocked by lack of receiving room. The s-flow is blocked when the r-reservoir is full and the S-flow when both are full.† (Note that it is possible for priority flow to advance when the regular flow is blocked, which is the advantage of the special lanes.)

This physical model can be used with more complex vehicle mixtures (e.g. differentiated by destination for multicommodity traffic modeling over networks) and at locations where the number of special lanes changes.

It is convenient to keep track of the receiving reservoirs status by means of a 'status box' diagram; see Fig. 6b. The status box is the rectangle defined by the two flow coordinate axes and the two perpendicular lines passing through the receiving point (R,r). The dots scattered about the figure represent possible locations of the sending point (S,s). The arrows point to the companion points whose coordinates are the final flows transferred by vehicle type; i.e. the IT flows (Q,q).

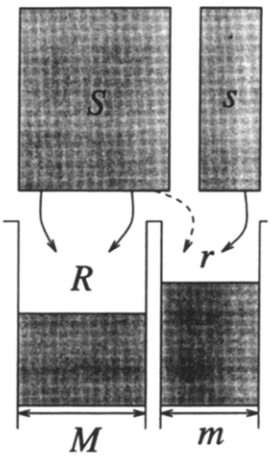
To understand why this is true it helps to visualize the cumulative fluid quantities transferred by a 'status point' (not shown in Fig. 6b to avoid clutter) that moves along the segment connecting the origin and the proper (S,s). As long as the slope of the line connecting the

*Although their origin is unknown to these authors, reservoir metaphors to describe freeway flow have been around for years. The particular form about to be discussed is significant because it yields the Godunov flows.

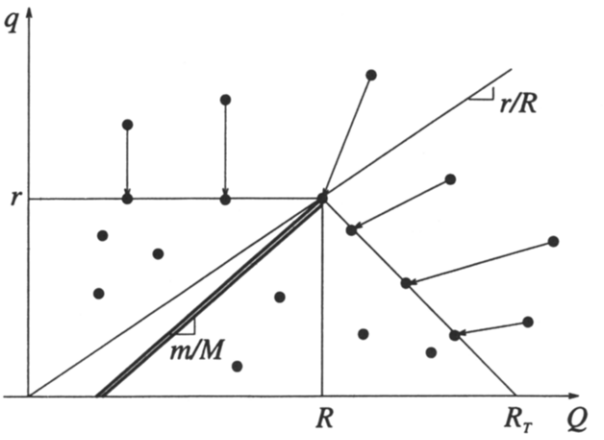
†The IT mechanism is a generalization of the gradual transfer model for diverges described in Daganzo (1995a).

status point and (R,r) (i.e. the ratio of the currently available spaces in the r- and R-reservoirs) is less than m/M the current height of the R-reservoir remains below that of the r-reservoir and each fluid goes into its separate reservoir. This occurs if the status point on the diagram is on the left side of the slanted double segment that stretches from the Q-axis to point (R,r) in the figure. On the right of this barrier flows are mixed, with priority vehicles moving into both reservoirs. We call this segment a 'mixing barrier'. Note that the mixing barrier cannot intersect the Q-axis to the left of the origin because $r/R \leqslant m/M$.

It should be clear that if the segment between (S,s) and the origin intersects the mixing barrier, the incremental amounts of transferred fluids will be represented by a succession of points on this line. Before the point of intersection the two vehicle types move into separate reservoirs, since the R-reservoir is filled to a lesser height. At that point, however, the two reservoir levels become equal and from then on the priority vehicles must advance into both reservoirs so as to keep the two reservoir levels even. This means that both reservoirs would reach capacity simultaneously (if at all). As a result we see that (Q,q) must equal (S,s), if (S,s) is to the left of the $-45°$ degree line passing through (R,r), and that, otherwise, (Q,q) will be at the point of intersection of this line with the segment. This result is illustrated by some of the arrows in the figure. Similar considerations reveal that if the segment joining (S,s) and the origin does not intersect the mixing barrier, then the (Q,q) points are as identified by the arrows.



(a) reservoir representation



(b) the receiving box diagram

Fig. 6. The incremental transfer principle.

The IT results described by Fig. 6b can be summarized by the following simple recipe:

$$q = \min\{s, r\} \text{ and } Q = \min\{S, R\}, \text{ if } S/s \leqslant R/r \tag{14a}$$

$$Q + q = \min\{S + s, R + r\} \text{ and } Q/q = S/s, \text{ if } S/s \geqslant R/r \tag{14b}$$

The latter equation can also be written as follows:

$$\left[\begin{array}{l} q = \min\{s, s(R+r)/(S+s)\} \\ Q = \min\{S, S(R+r)/(S+s)\} \end{array}\right] \text{ if } S/s \geqslant R/r. \tag{14c}$$

The next section shows that the incremental transfer principle indeed yields the Godunov flows eqns (7)–(11).

3. EQUIVALENCE OF THE INCREMENTAL TRANSFER FLOWS AND THE GODUNOV FLOWS

Section 3.1 below demonstrates the equivalence analytically, and Section 3.2 provides numerical evidence. Section 3.1 may be skipped without loss of continuity.

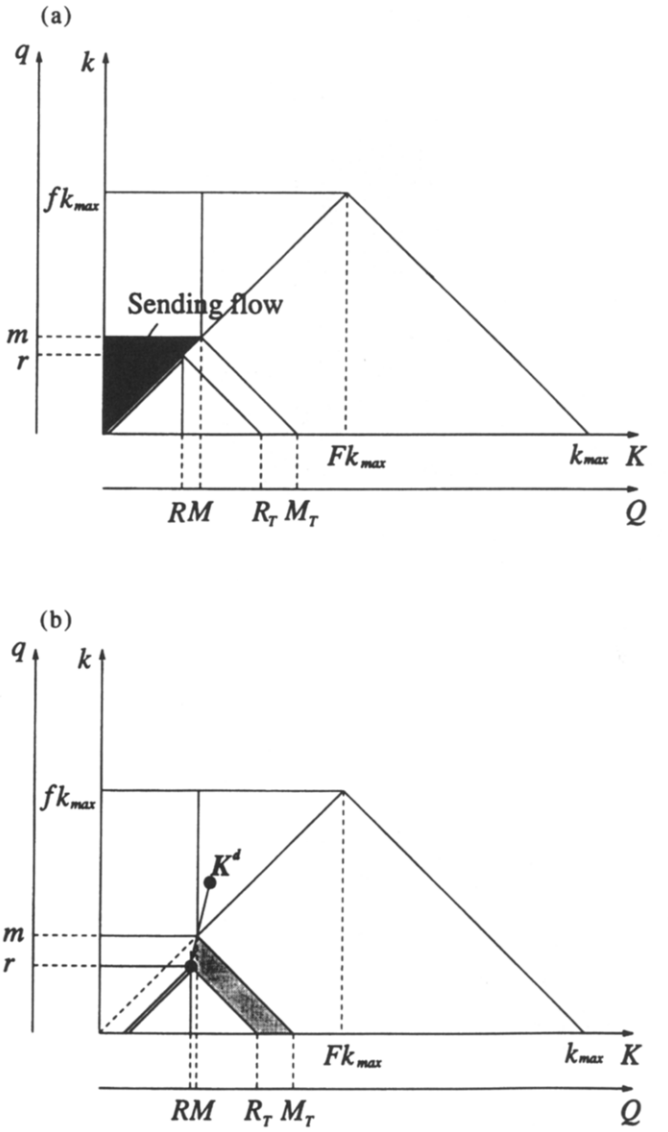


Fig. 7(a) and (b) (caption on page 113).

3.1. Analytical Proof

This is done by considering the 4 possible cases that can arise, depending on whether the upstream and downstream states are in 1-pipe or 2-pipe regimes.

3.1.1. 1-pipe to 1-pipe

In this case the Godunov flows are given by eqn (11) and the input values of the proposed recipe (S, s, R and r) by eqn (12). Since $S/s \geqslant M/m = R/r$, the proposed flows are then given by eqn (14b). The original and proposed results are equivalent, because eqn (11) states the same formula as eqns (12) and (14b).

3.1.2. 2-pipe to 2-pipe

In this case the Godunov flows are given by eqn (9) and the input values of the proposed recipe by eqn (13). From Figs 4 and 5 we also know that $S/s \leqslant M/m \leqslant R/r$, so that the proposed flows are given by eqn (14a). Again, the original and proposed results are equivalent because eqn (13) and eqn (14a) restate eqn (9).

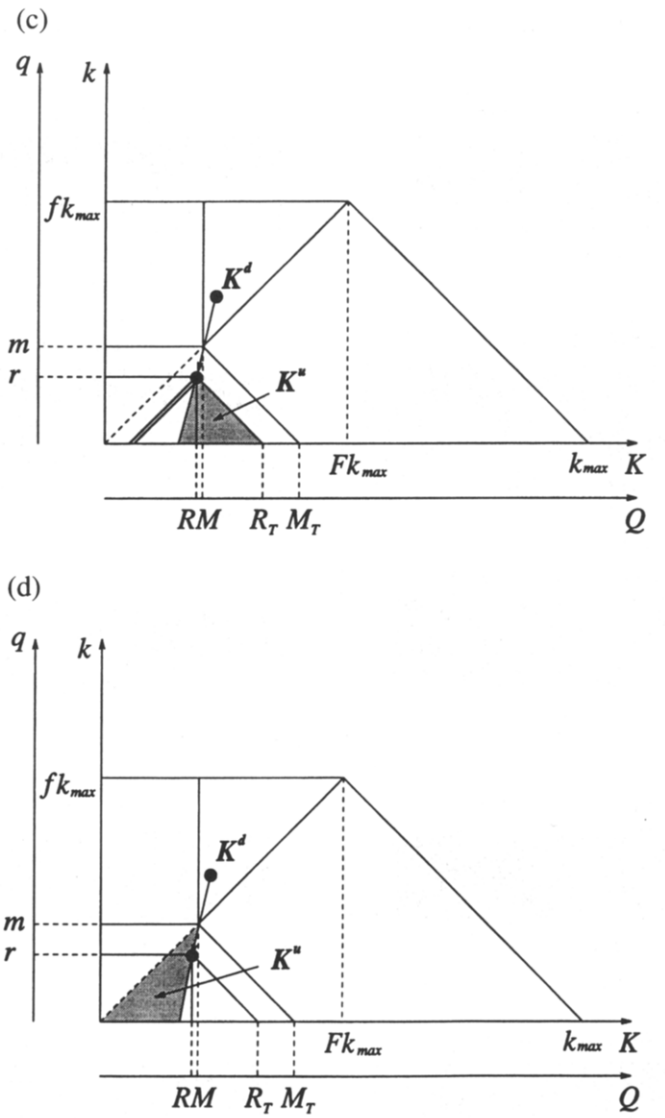


Fig. 7. Receiving box diagram for two kinds of Riemann problems: (a) 2-pipe to 1-pipe; (b) 1-pipe to 2-pipe [upstream congested or in subcase 1 with $S_T(K_T^u) > r(k^d) + R(K^d)$], (c) 1-pipe to 2-pipe in subcase 1 with $S_T(K_T^u) > r(k^d) + R(K^d)$] and (d) 1-pipe to 2-pipe in subcase 2.

3.1.3. 2-pipe to 1-pipe

In this case the Godunov flows are given by eqn (10) and the input values of the proposed recipe by eqns (12b) and (13a). From Figs 4 and 5 we also know that $S/s \leqslant M/m = R/r$, so that the receiving box is as in Fig. 7a, with the mixing barrier passing through the origin and the sending points in the shaded area. This means that the proposed flows are given by eqn (14a).

If $\mathbf{K}^d$ is in regime 'A' we see from eqn (12b) that $(R, r) = (M, m)$. Thus, eqn (14a) reduces to $(Q, q) = (S, s)$, which is equivalent to eqn (10(a,b)). Conversely, if $\mathbf{K}^d$ is in regime 'D', we see from eqn (12b) and Fig. 5 that

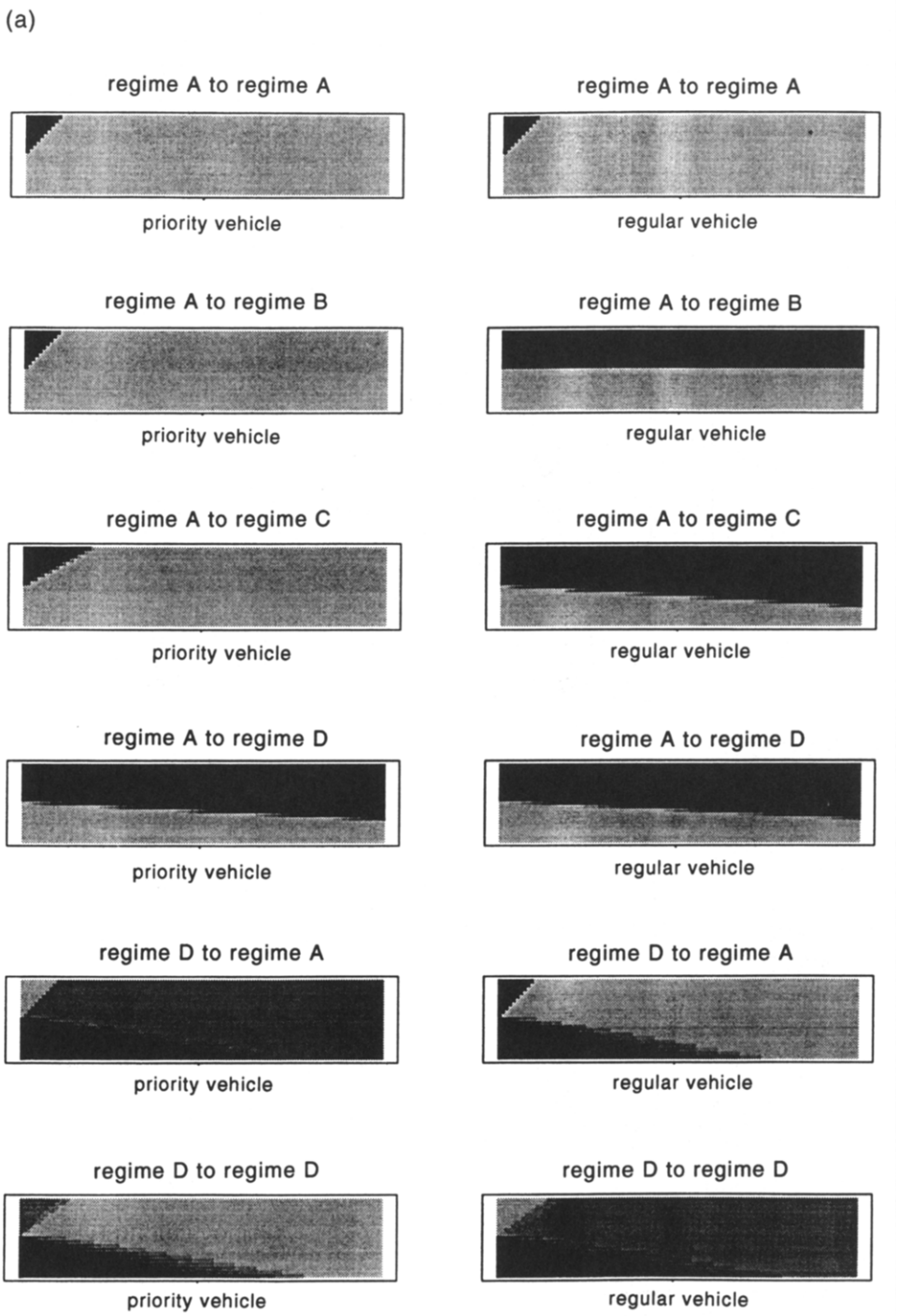


Fig. 8(a)—*Continued opposite* (caption on page 117).

$$R = R_T(K+k)M/(M+m) \equiv R((K+k)M/(M+m))$$

and

$$r = R_T(K+k)m/(M+m) \equiv r((K+k)m/(M+m)),$$

where the identities come from the definition of the functions $R()$ and $r()$ given in Section 1.2. Clearly, then eqn (14a) expresses the same relations as eqn (10b).

(b)

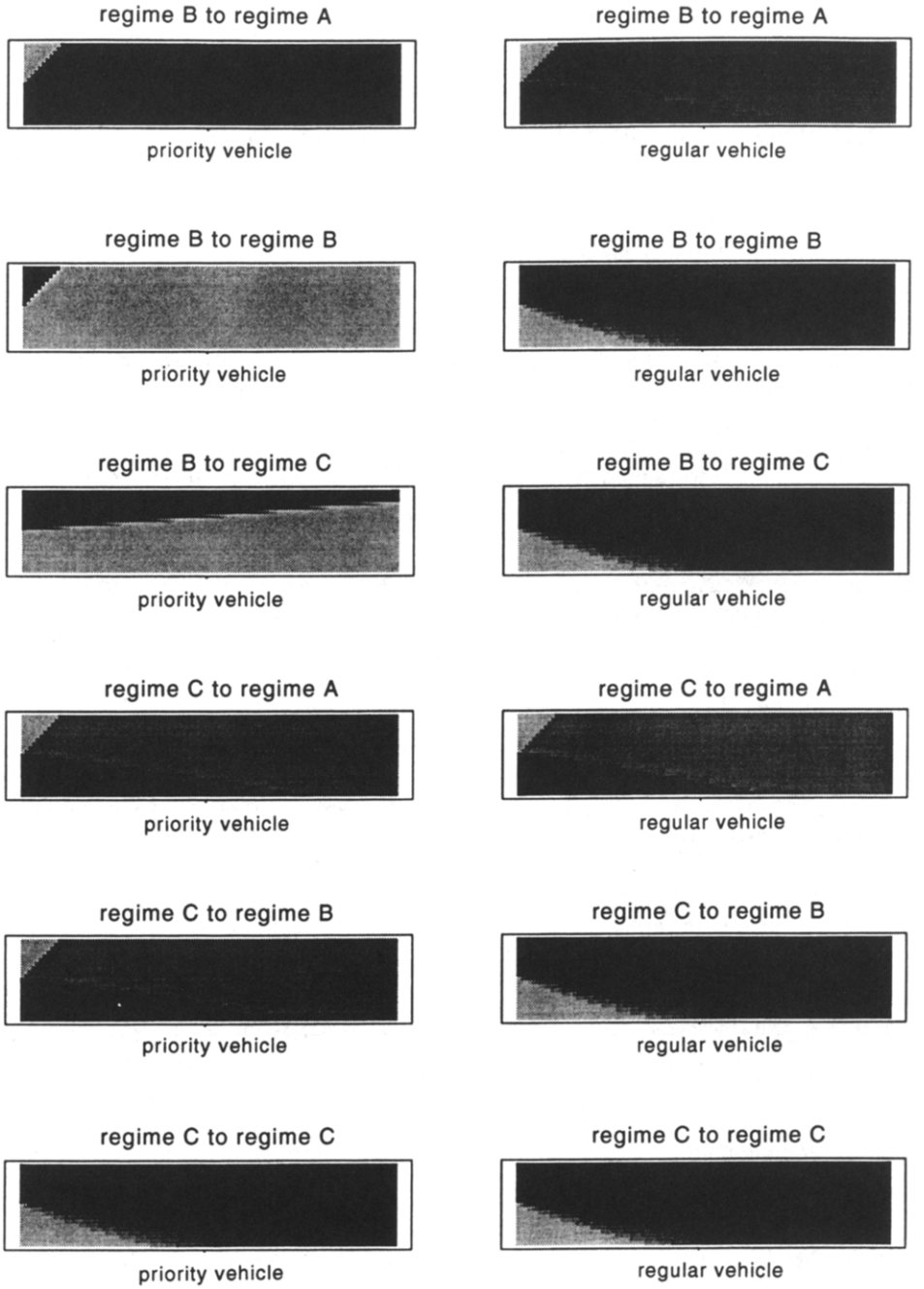


Fig. 8(b)—*Continued overleaf* (caption on page 117).

3.1.4. 1-pipe to 2-pipe

In this case three possible cases can arise.

3.1.4.1. Case (i) — Godunov flows given by eqn (7). We have seen that this case arises if the downstream state is congested or semi-congested, and the upstream state is either in congested region 'D' or in (uncongested) subcase 1 with $S_T(K_T^u) > r(k^d) + R(K^d)$.

If the downstream state is congested as in Fig. 7b, then the receiving point, whose coordinates are $(R, r) = (R(K^d), r(k^d))$ by virtue of eqn (13b), is located where indicated by the construction of Fig. 5a with $(R, r) < (M, m)$. It is also true that the sending point

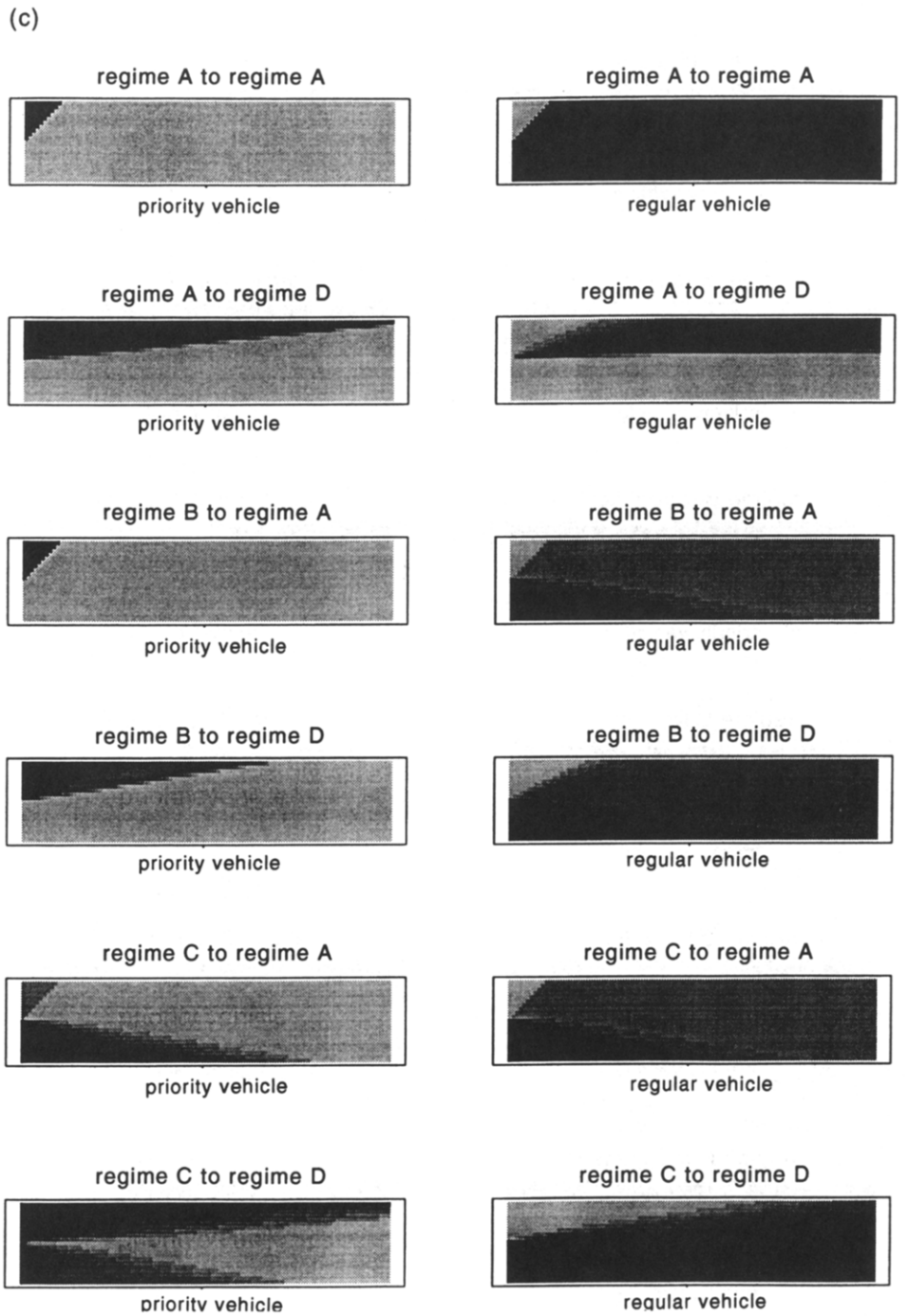


Fig. 8(c)—*Continued opposite* (caption on page 117).

must be in the shaded area of Fig. 7b. Then, application of the IT recipe embodied in Fig. 6b to the case in our figure reveals that:

$$Q + q = R + r \text{ and that } q = \min\{r, (R + r)s/(S + s)\}; \tag{15}$$

and since $s/(S + s) = k^u/(K^u + k^u)$ in the 1-pipe regime, we see that eqn (15) is equivalent to eqn (7a and c).

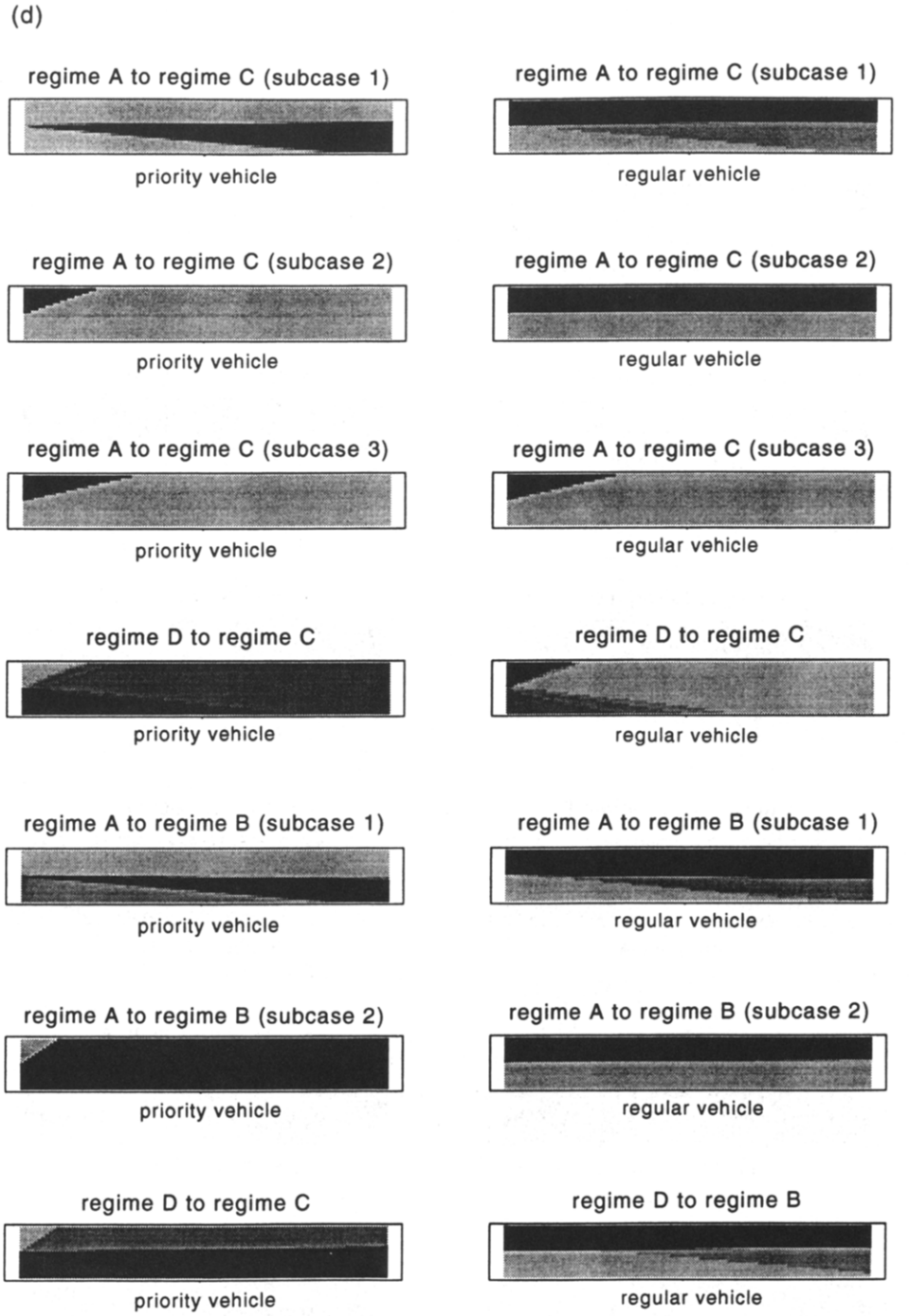


Fig. 8. Density plots with IT principle. Four cases: (a) no boundary crossings, with K^u in regions A or D; (b) no boundary crossings, with K^u in regions B or C; (c) boundary crossing of the 'lead-vehicle' type, with K^u in 2-pipe regime and K^d in regions A or D; and (d) boundary crossing of the 'moving-bottleneck' type, with K^u in regions A or D and K^d in regions B or C.

The same result applies if the downstream state is semi-congested, as can be seen by repeating the above steps with the appropriately modified receiving box.

3.1.4.2. Case (ii) — Godunov flows given by eqn (8). This case arises if $\mathbf{K}^u$ is in subcase 1 and $S_T(K_T^u) \leqslant r(k^d) + R(K^d)$. These conditions mean that $\mathbf{K}^u$ and the sending point coincide and are in the shaded region of Fig. 7c. Clearly then, application of the IT principle yields $(Q, q) = (S, s)$, which is the same result as eqn (8).

The same can be said if the downstream state is uncongested or semi-congested.

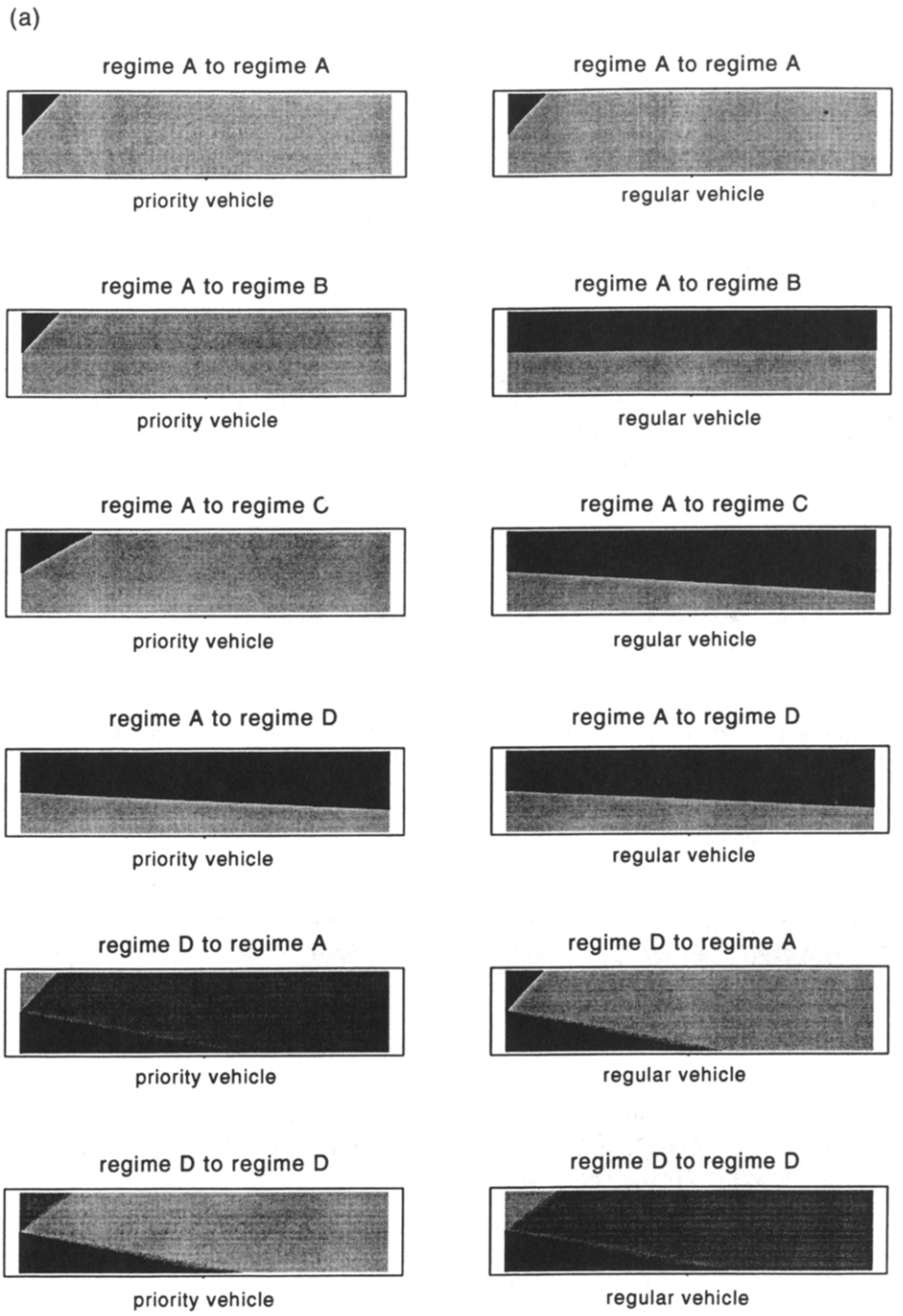


Fig. 9(a)—*Continued opposite* (caption on p. 121).

3.1.4.3. Case (iii) — Godunov flows given by eqn (9). This arises if $\mathbf{K}^u$ is in the shaded region of Fig. 7d. Then the sending point is on the same spot in the diagram (see Fig. 4) and it becomes clear from Fig. 6b that the (Q,q) obtained from the IT principle matches eqn (9(a,b)), as claimed.

The case where the downstream state is either uncongested or semi-congested can be treated similarly with the same result.

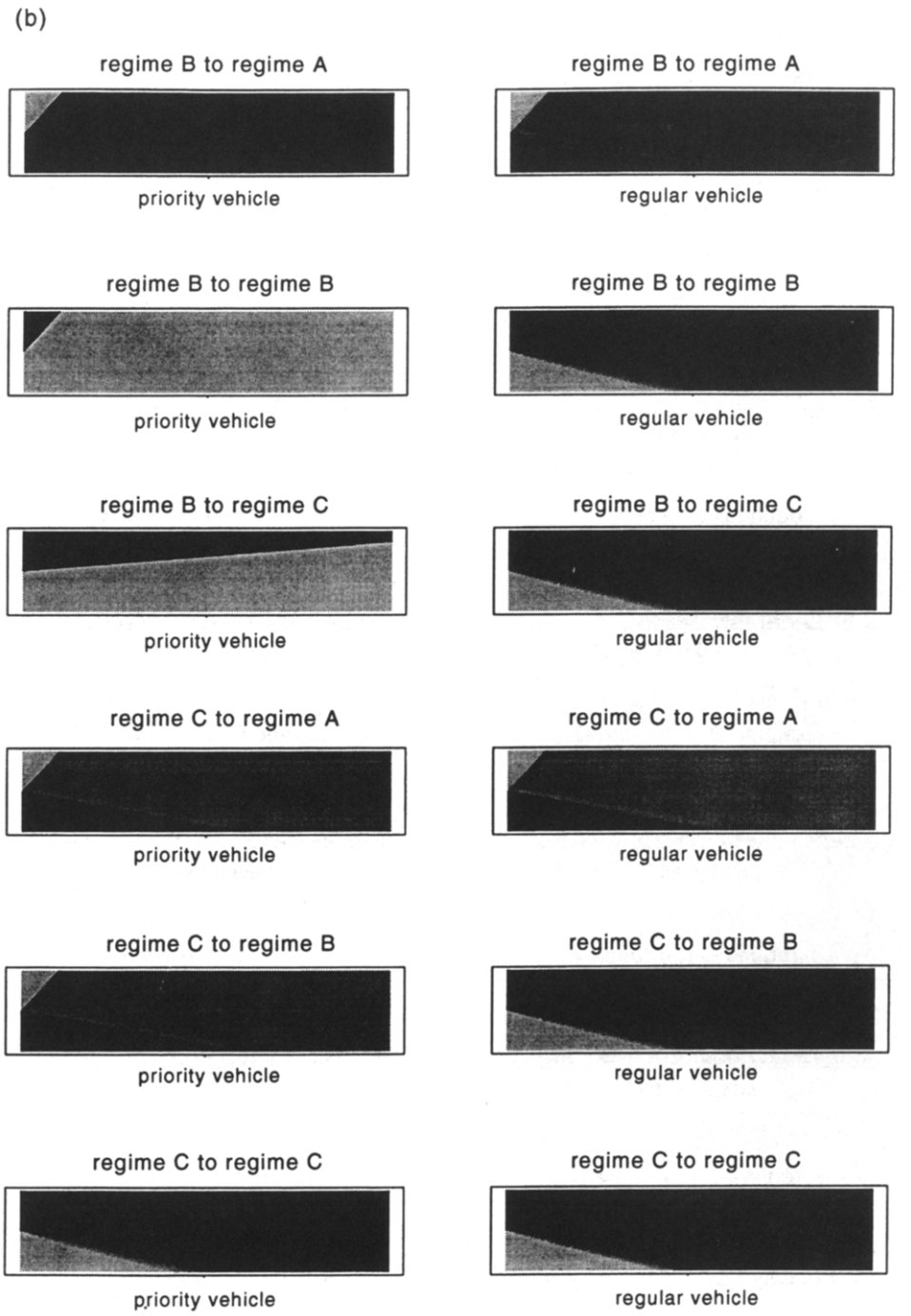


Fig. 9(b)—*Continued overleaf* (caption on page 121).

3.1.4.4. Case (iv) — Godunov flows given by eqn (11). This case arises if the downstream state is uncongested. Note that eqn (11) reduces then to a form that is independent of $\mathbf{K}^d$; i.e.:

$$(Q, q) = (K^u/K_T^u, k^u/K_T^u)S_T(K^u + k^u).$$

The receiving flows of the IT procedure are (M,m); i.e. the same as if the downstream state had been in the uncongested region of the 1-pipe regime. Thus, the result of the IT procedure is the same as for a 1-pipe to 1-pipe transition into an uncongested downstream

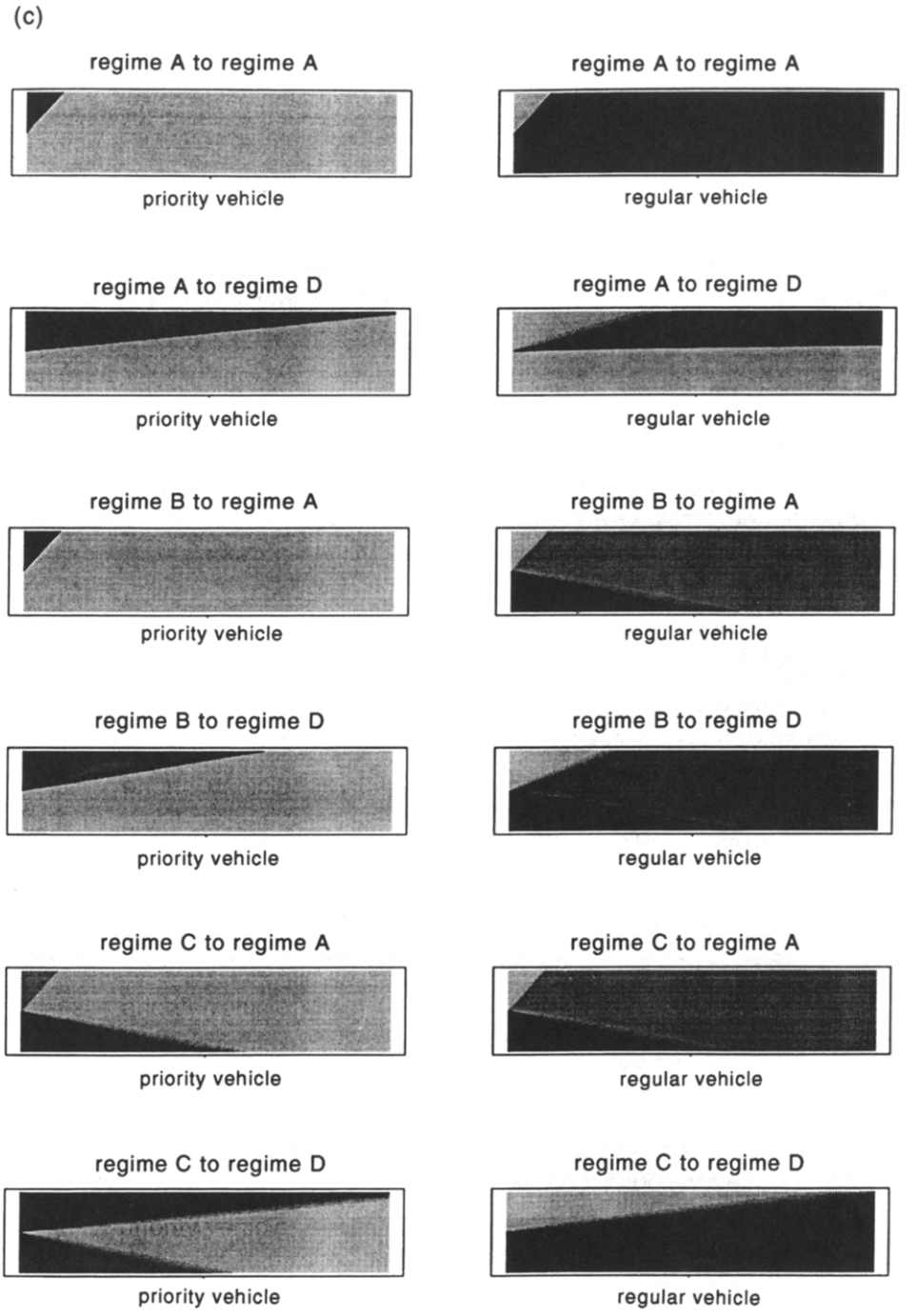


Fig. 9(c)—*Continued opposite* (caption on page 121).

state, which we have already shown to be equivalent to eqn (11). Since neither eqn (11) nor the IT procedure are affected by the specific downstream state used we see that the results of both recipes must match.

We have now considered all possible cases and have established the desired equivalences in every instance. Therefore our claim is true. The next subsection demonstrates the result numerically for all possible Riemann problems.

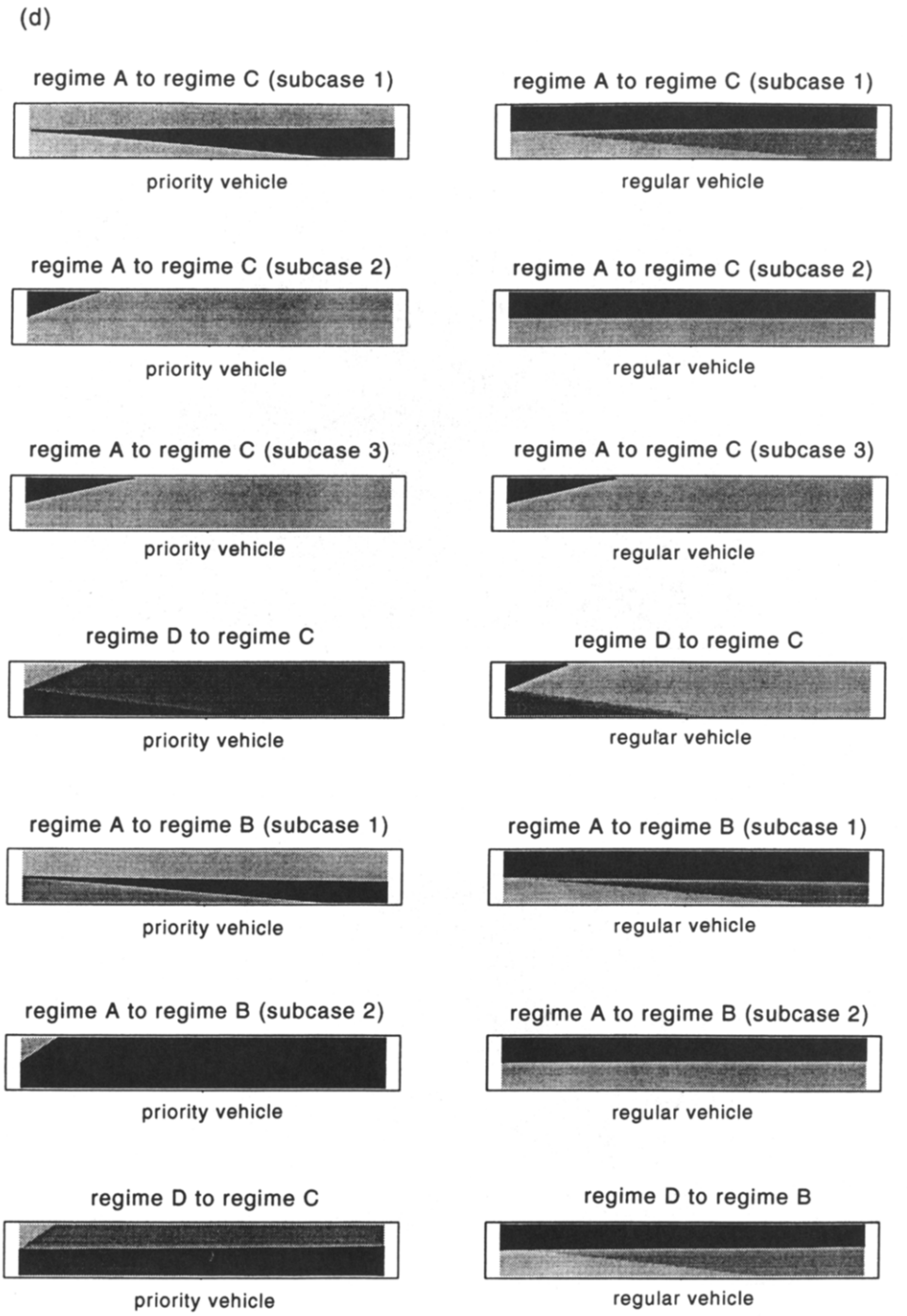


Fig. 9. Density plots with IT principle (5 times higher resolution than in Fig. 8). Same four cases: (a) no boundary crossings, with K^u in regions A or D; (b) no boundary crossings, with K^u in regions B or C; (c) boundary crossing of the 'lead-vehicle' type, with K^u in 2-pipe regime and K^d in regions A or D; and (d) boundary crossing of the 'moving-bottleneck' type, with K^u in regions A or D and K^d in regions B or C.

3.2. Numerical Demonstration

Figures 5–7 of Daganzo (1997), display the solution to every possible Riemann problem. Likewise, Fig. 8(a-d) displays the results obtained with the IT principle, eqns (12–14), for the same collection of problems. Each case in Fig. 8 is shown by a pair of (t,x) intensity plots in which the density of shading is proportional to the vehicular density; the left plot gives $K(t,x)$ and the right plot $k(t,x)$.* The reader can verify that in every case the interfaces between areas of different shading intensity slant with the slope indicated by the theory.

The fuzzy boundary between areas is caused by numerical error ('numerical diffusion') and its effect can be minimized by reducing ϵ and δ. This is shown in Fig. 9(a–d), which

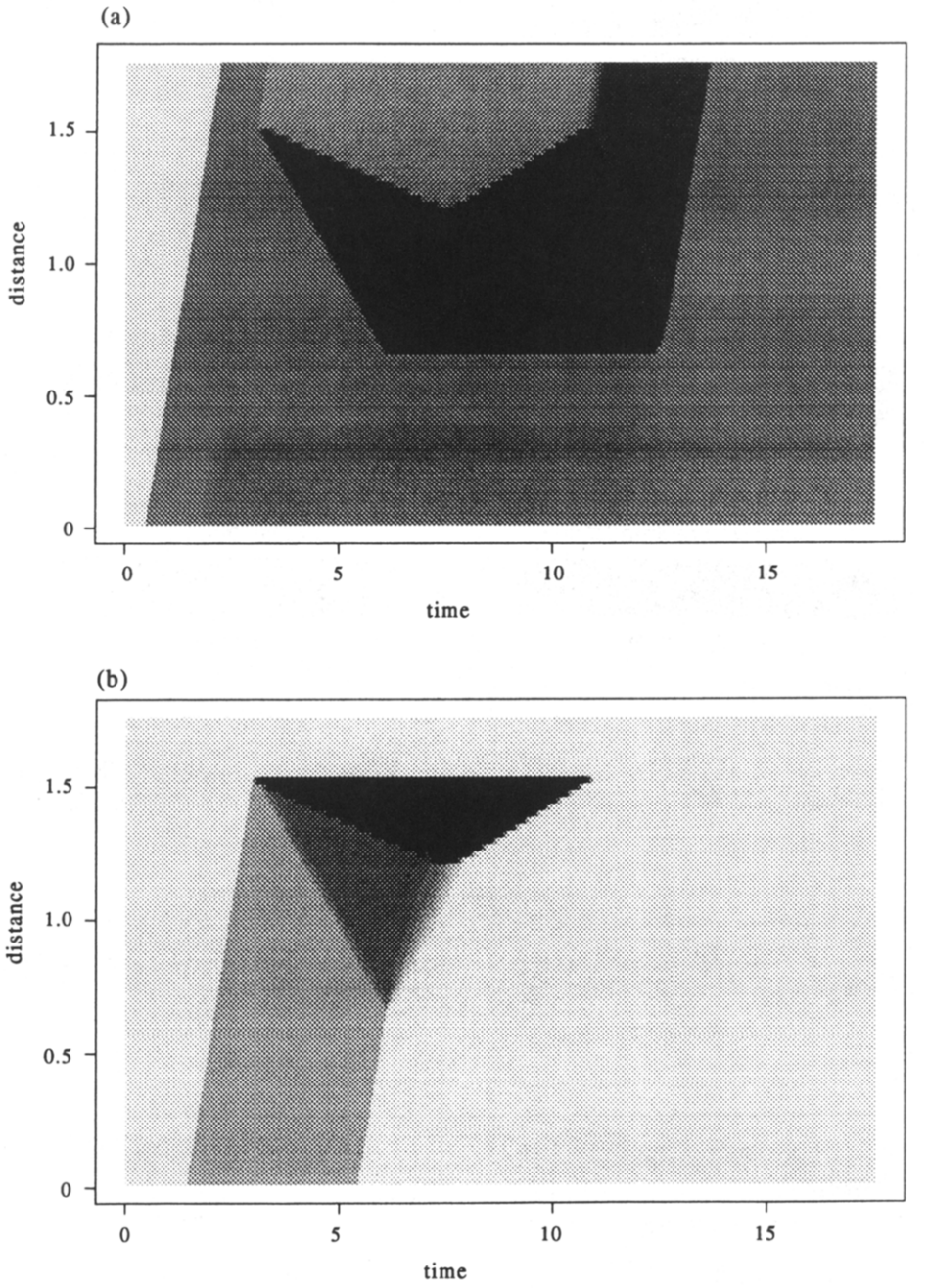


Fig. 10. Density map for a problem involving an off-ramp spillover.

*Shading intensities cannot be compared across plots because the proportionality constant varies across plots.

display all the cases of Fig. 8 with five times smaller (ϵ,δ). Formulae predicting the extent of numerical diffusion can be found in Daganzo (1995b).

In the process of preparing Fig. 8 we also verified that the IT formulas were equivalent to the Godunov flows, as shown in Section 3.1. This is important because it confirms that no algebraic errors were made in the derivation of the original and proposed formulae. The numerical comparisons of Figs 8 and 9 are also important because they show that the procedure converges for the Riemann problems. (Godunov's method, although successful in many practical applications has not been proven to converge for the kind of problem that concerns us.) Therefore, since most practical problems are formulated as a sequence of Riemann problems, the numerical results further establish the good convergence properties of the proposed method in practical cases.

4. THE INHOMOGENEOUS FREEWAY, BOUNDARY CONDITIONS AND AN EXAMPLE

The incremental transfer principle also suggests a way in which a discrete change in the number of lanes allocated to one or both of the vehicle types can be modeled. It seems reasonable to use eqns (12) and (13) to define the 4 sending and receiving values for every cell (on both sides of the change) and then use the principle to determine the flows that can advance across the discontinuity. It is beyond the scope of this paper to discuss the mathematical implications of this statement, but it should be reassuring to see that nothing in the IT principle [eqn (14)] uses the lane configuration of the upstream cell.

As a preliminary exploration of these ideas we revisit the example in Section 5.3 of Daganzo (1997). This consisted of a congested exit ramp. Upstream of the ramp the freeway is modeled as one with special lanes, $(F,f)=(0.5, 0.5)$, where the exiting vehicles are 'regular' and must stay on the 'near' side (regular) lanes. The intersection can be modeled as a reservoir that flows into two downstream ones (the exit ramp and the through lanes), with the rule that priority vehicles must go in the 'through' reservoir and regular vehicles in the 'exit' reservoir. No vehicles have a choice of reservoir, and each flow is blocked when its reservoir is full. Conceivably, one could connect these downstream cells to others and model in such a way a network. For the current example, however, this is not necessary; it suffices to assume that the downstream reservoirs emit all their contents at each iteration.

Figure 10 displays the density plot obtained with the method just described, using 700 time steps ($\epsilon = 0.025$) and $\delta/\epsilon = v_f$, which matches rather well the exact solution given in Fig. 10c of Daganzo (1997).

A more quantitative comparison is made in Fig. 11 by superimposing the exact (solid lines) and approximate (dashed lines) cumulative curves of vehicle counts (Newell curves) at various locations (also for $\epsilon=0.025$). The match improves with decreasing lattice sizes; in our case, the two sets of curves became indistinguishable when the mesh was reduced in scale 15 times. As is well known, the N-curves give, at a glance, the vehicular trip times and the accumulation between observation points; therefore, the discrepancies shown in the figure have a clear physical interpretation. The good fit indicates that the exact solution of the example can be approximated as closely as desired with an IT simulation.

Numerical experimentation also reveals that the IT principle approximates as well as desired the physically relevant solutions of the Riemann initial value problems for our junction — where the downstream leg of of the freeway is assumed to remain uncongested.*

This is encouraging because it suggests that the principle may be rather general.

*Physical considerations suggest that traffic directly upstream of our junction can only be found in one of the following three stable states: (i) the ramp is undersaturated with no queue and there is a free-flow state upstream of the junction, (ii) the ramp is saturated by a queue that fills the exit lanes and there is a semi-congested state directly upstream of the junction, or (iii) there is a barely emerging queue that partially blocks the exit lanes and generates a 1-pipe, congested state directly upstream of the junction. Possibility (iii) has been noted by G.F. Newell. It turns out that every Riemann problem for our junction has a unique solution that belongs to one and only one of these categories. We have found that the IT rule solves the Riemann problem exactly in cases (i) and (ii), and approximately (but as accurately as desired in terms of the cumulative flows) in case (iii).

5. DISCUSSION

This paper has shown that Godunov's finite difference approximation method for the problem of kinematic waves on freeways with special lanes reduces to an intuitive physical idea that we call the incremental transfer (IT) principle. Numerical examples constructed for the various Riemann problems that can arise indicate that the IT/Godunov approach converges to the exact solution as the mesh size is decreased. We caution, however, that other seemingly intuitive principles we checked are not equivalent to Godunov's approach and appeared not to converge in the proper way.

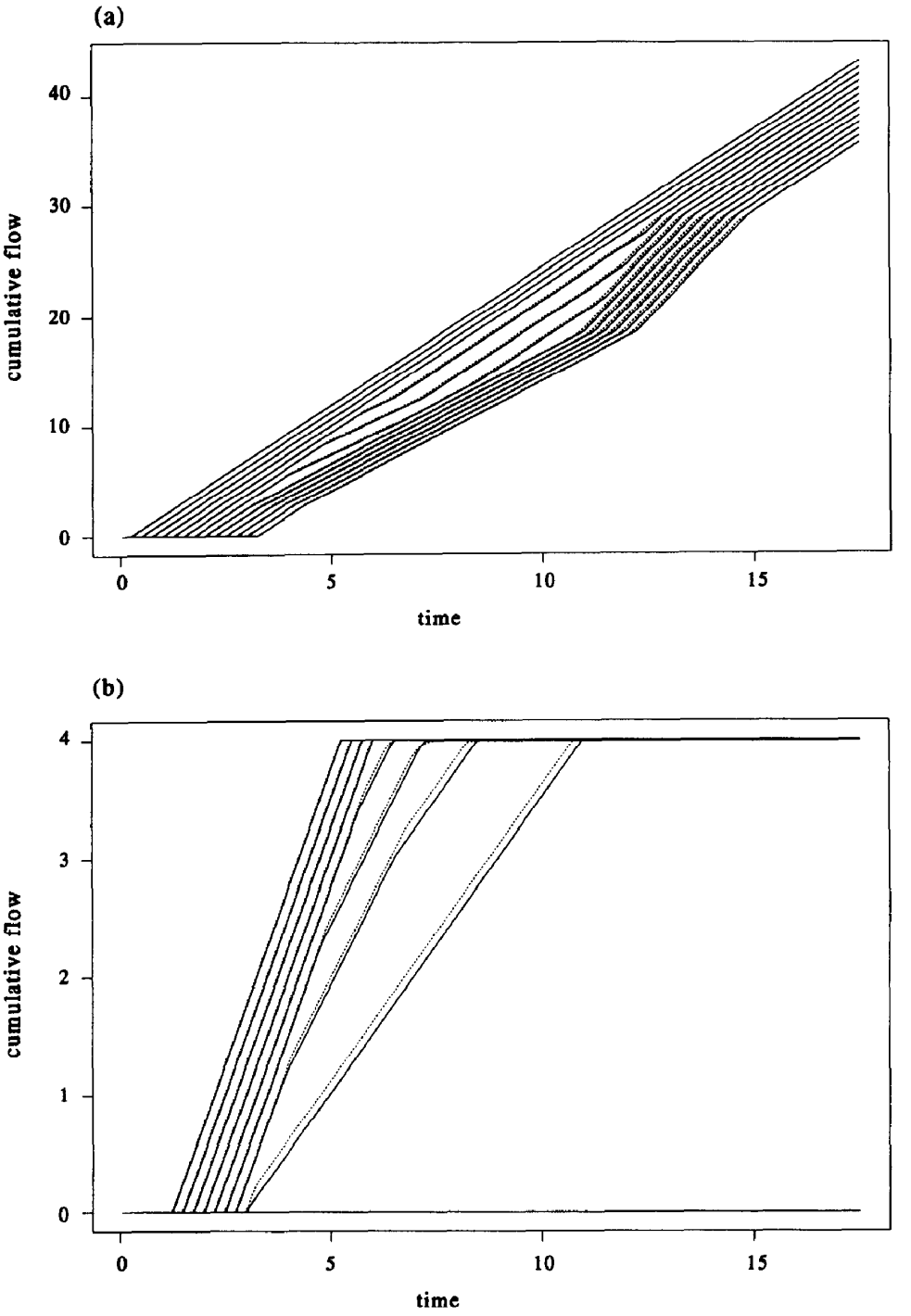


Fig. 11. Exact and approximate N-curves for the ramp spillover problem.

The new theory has the potential for depicting more realistically the formation and dissipation of queues upstream of a diverge junction than other existing traffic models. In some models, when the flow into one of the diverge branches is blocked, the flow into the other branch is reduced simultaneously; i.e. contrary to observation, a coalesced queue including both vehicle types is assumed to form immediately upstream of the junction area. Exceptions to this can be observed regularly in the San Francisco/Oakland/Berkeley area; e.g. on Westbound Hwy 24, upstream of the Walnut Creek interchange during the evening peak, on Northbound Hwy 13, upstream of the interchange with Hwy 24 during the evening peak, and on Southbound Hwy 101, directly upstream of the Candlestick (now 3Com) Stadium exit on game days. In the proposed model, the queue formed by the exiting vehicles behaves like a bottleneck whose upstream end may move in the upstream or downstream direction. If through vehicles arrive in significant numbers to this bottleneck they may also have to queue upstream of it (intermingled with turning vehicles) but they would flow freely from that point onwards. The next step is to see whether these predictions of the simple kinematic wave theory are also observed in detector data; the IT principle should facilitate the task.

Of course, the results of this paper also apply to other situations with special lanes and two vehicle types, such as freeways with carpool lanes. We also conjecture that the IT principle may be appropriate to describe inhomogeneous highways, networks with junctions, and road systems with more complicated lane configurations; e.g. involving any number of vehicle types (i), lane types (j), and a set of variables δ_{ij} that would indicate whether i-vehicles are allowed in the j-lanes or not. More research is needed to determine whether these conjectures are true, however. If so, it would be a simple matter to incorporate the proposed ideas into network models.

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